

Lecture Notes on Thermodynamic Computing

Chapter 1: Fundamentals of Thermodynamic Computing Devices

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1 1.1 Introduction to Thermodynamic Computing

• What is Thermodynamic Computing?

Definition 1.1 (Thermodynamic Computing). Thermodynamic computing refers to computational systems that harness thermal noise and statistical fluctuations as fundamental computing resources, rather than treating them as unwanted disturbances.

Key characteristics:

- Utilizes thermal fluctuations at scale $k_B T$
- Operates near the thermodynamic limit
- Probabilistic rather than deterministic computation
- Ultra-low power consumption

• Motivation for Thermodynamic Computing

Energy Crisis in CMOS:

- Moore's Law approaching physical limits
- Power density wall: $> 100 \text{ W/cm}^2$
- Battery limitations in mobile devices
- Heat dissipation challenges

Fundamental Energy Limits:

$$E_{switching} = \frac{1}{2} C V_{DD}^2 \gg k_B T \ln 2 \quad (1)$$

Current gap: 10^4 to 10^6 times above thermodynamic limit

¹Figures will be corrected by 10/30/2025

2 1.2 Physical Principles

• Thermal Voltage

The thermal voltage at temperature T :

$$V_T = \frac{k_B T}{q} \approx 26 \text{ mV at } 300\text{K} \quad (2)$$

This sets the natural voltage scale for thermodynamic computing.

• Boltzmann Statistics in Devices

For electrons in thermal equilibrium:

$$n(E) = n_0 \exp\left(-\frac{E - E_F}{k_B T}\right) \quad (3)$$

Subthreshold current in MOSFETs:

$$I_{ds} = I_0 \exp\left(\frac{V_{gs} - V_{th}}{nV_T}\right) \quad (4)$$

• Fluctuation-Dissipation Theorem

Johnson-Nyquist noise:

$$\langle V_n^2 \rangle = 4k_B T R \Delta f \quad (5)$$

Shot noise:

$$\langle I_n^2 \rangle = 2q I_{dc} \Delta f \quad (6)$$

3 1.3 Thermodynamic Computing Paradigms

• Stochastic Computing

Definition 3.1 (Stochastic Computing). Information encoded in the probability of random bit streams:

$$P(X = 1) = x \quad \text{represents value } x \in [0, 1] \quad (7)$$

Operations:

- Multiplication: AND gate
- Addition: OR gate (scaled)
- Complex functions via polynomial expansion

• Probabilistic Computing

Based on Bayes' theorem:

$$P(H|E) = \frac{P(E|H)P(H)}{P(E)} \quad (8)$$

Applications:

- Machine learning inference
- Signal processing under uncertainty
- Optimization problems

4 1.4 Device Physics of Thermodynamic Computing

• Subthreshold Operation

Operating transistors below threshold ($V_{gs} < V_{th}$):

Advantages:

- Ultra-low power: $I_{ds} \sim \text{nA to pA}$
- Exponential I - V characteristics
- Natural access to thermal noise

Challenges:

- Process variations become significant
- Slow switching speeds
- Increased susceptibility to noise

• Thermal Noise as Signal

Signal-to-noise ratio in subthreshold:

$$SNR = \frac{I_{signal}^2}{\langle I_{noise}^2 \rangle} = \frac{g_m^2 V_{in}^2}{2qI_{ds}\Delta f} \quad (9)$$

Where transconductance:

$$g_m = \frac{I_{ds}}{nV_T} \quad (10)$$

5 1.5 Energy Efficiency Analysis

• Switching Energy

Dynamic energy per switching event:

$$E_{dyn} = \frac{1}{2}C_{load}V_{DD}^2 \quad (11)$$

Static energy per operation:

$$E_{static} = V_{DD}I_{leak}t_{prop} \quad (12)$$

- **Subthreshold Energy Optimization**

Total energy:

$$E_{total} = \frac{1}{2}C_{load}V_{DD}^2 + V_{DD}I_{off}t_{prop} \quad (13)$$

Optimal supply voltage:

$$V_{DD}^{opt} = 2nV_T \ln \left(\frac{C_{load}}{I_0 t_{prop} n V_T} \right) \quad (14)$$

- **Energy-Delay Product**

For subthreshold circuits:

$$EDP = CV_{DD}^2 \cdot \frac{CV_{DD}}{I_{ds}} \propto V_{DD}^3 \exp \left(-\frac{V_{DD}}{nV_T} \right) \quad (15)$$

Minimum occurs near $V_{DD} = 3nV_T$

6 1.6 Noise-Tolerant Computing Architectures

- **Redundancy and Error Correction**

Triple modular redundancy:

- Compute same operation 3 times
- Majority voting for output
- Energy overhead: $3 \times$ computation + voting

- **Probabilistic Error Models**

Bit error probability in subthreshold:

$$P_e = \frac{1}{2} \exp \left(-\frac{V_{signal}^2}{2\sigma_{noise}^2} \right) \quad (16)$$

Where noise variance:

$$\sigma_{noise}^2 = \frac{k_B T}{C_{total}} \quad (17)$$

7 1.7 Applications of Thermodynamic Computing

- **Bayesian Neural Networks**

Synaptic weights as probability distributions:

$$w \sim \mathcal{N}(\mu_w, \sigma_w^2) \quad (18)$$

Forward pass with uncertainty:

$$y = f(x) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma_y^2) \quad (19)$$

- **Optimization and Sampling**

Simulated annealing implementation:

$$P_{accept} = \exp\left(-\frac{\Delta E}{k_B T_{eff}}\right) \quad (20)$$

Where effective temperature can be controlled electrically.

- **Sensor Fusion**

Combining noisy measurements:

$$x_{fused} = \frac{\sum_i w_i x_i}{\sum_i w_i}, \quad w_i = \frac{1}{\sigma_i^2} \quad (21)$$

Natural implementation in subthreshold circuits.

8 Examples

Example 8.1 (Subthreshold Inverter Energy). Calculate energy per operation for subthreshold inverter.

Given:

- $V_{DD} = 0.2$ V
- $C_{load} = 1$ fF
- $I_{off} = 1$ pA
- $t_{prop} = 1$ s

Solution:

Dynamic energy:

$$E_{dyn} = \frac{1}{2} \times 10^{-15} \times (0.2)^2 \quad (22)$$

$$= 2 \times 10^{-17} \text{ J} = 0.125 \text{ aJ} \quad (23)$$

Static energy:

$$E_{static} = 0.2 \times 10^{-12} \times 10^{-6} \quad (24)$$

$$= 2 \times 10^{-19} \text{ J} = 0.00125 \text{ aJ} \quad (25)$$

Total: $E_{total} = 0.126 \text{ aJ} \approx 30k_B T$

Example 8.2 (Thermal Noise Analysis). Find signal-to-noise ratio for subthreshold amplifier.

Given:

- $I_{ds} = 10$ nA
- $V_{in} = 10$ mV
- $\Delta f = 1$ MHz

- $n = 1.3$

Solution:

Transconductance:

$$g_m = \frac{I_{ds}}{nV_T} = \frac{10 \times 10^{-9}}{1.3 \times 26 \times 10^{-3}} \quad (26)$$

$$= 296 \text{ nS} \quad (27)$$

Signal power:

$$P_{signal} = (g_m V_{in})^2 R_{load} \quad (28)$$

$$= (296 \times 10^{-9} \times 10 \times 10^{-3})^2 \times 10^6 \quad (29)$$

$$= 8.76 \times 10^{-12} \text{ W} \quad (30)$$

Noise power:

$$P_{noise} = 2qI_{ds}\Delta f \times R_{load} \quad (31)$$

$$= 2 \times 1.6 \times 10^{-19} \times 10 \times 10^{-9} \times 10^6 \times 10^6 \quad (32)$$

$$= 3.2 \times 10^{-15} \text{ W} \quad (33)$$

$$\text{SNR} = \frac{8.76}{3.2} = 2.74 = 4.4 \text{ dB}$$

9 Homework Problems

Assignment 1.1: Fundamentals of Thermodynamic Computing

1. Thermal Voltage Scaling:

- Calculate thermal voltage at 77K, 300K, and 400K
- Find optimal V_{DD} for subthreshold operation
- Analyze temperature effects on power consumption

2. Subthreshold Characteristics:

- Plot I_{ds} vs V_{gs} for $n = 1.2, 1.5, 2.0$
- Find subthreshold swing for each case
- Calculate transconductance efficiency g_m/I_{ds}

3. Energy-Delay Optimization:

- Derive optimal V_{DD} for minimum EDP
- Consider both dynamic and static energy
- Plot EDP vs V_{DD} for different load capacitances

4. Noise Analysis:

- Calculate thermal noise in 1 k resistor at 300K
- Find equivalent input noise for subthreshold amplifier

- Design noise-matched input stage
5. **Stochastic Computing:**
- Design stochastic multiplier using AND gates
 - Calculate accuracy vs bit stream length
 - Implement square root function
6. **Process Variation Effects:**
- Model V_{th} variations with $\sigma = 50$ mV
 - Calculate I_{ds} spread in subthreshold
 - Design compensation techniques
7. **Temperature Effects:**
- Analyze I_{ds} temperature dependence
 - Find zero-temperature-coefficient bias point
 - Design temperature-stable reference
8. **Power Analysis:**
- Compare superthreshold vs subthreshold power
 - Calculate energy per FLOP for both modes
 - Find break-even frequency

Next Chapter Preview: Chapter 2 will explore "kT-RAM and Thermal Memory Technologies" including design principles and implementation challenges.